

CHEMISTRY

ASSIGNMENT

NADEC BIJU
MECHANICAL-A
SEMESTER 1

PROPERTIES OF LUBRICANTS

There are many properties of lubricants that are useful to us in our day-to-day life. We will be discussing about a few of these in this assignment

- VISCOSITY
- VISCOSITY INDEX
- FLASPOINT
- FIRE POINT
- CLOUD POINT
- POUR POINT
- ANILINE POINT
- STEAM EMULSION NUMBER
- NEUTRALIZATION NUMBER
- SAPONIFICATION NUMBER
- IODINE VALUE
- CARBON RESIDUE

Viscosity

Viscosity changes in temperature, the viscosity of lubricating oil varies, the higher the temperature, the lower the viscosity and vice versa. The rate of variation of viscosity with temperature is different for different base oil or fluids. The rate at which the viscosity of an oil changes with temperature is measured by an empirical number, known as viscosity index. A relatively small change in viscosity with temperature is indicated by high viscosity index. whereas low viscosity index shows a relatively large change in viscosity with temperature. oils of gulf coast region are arbitral assigned a viscosity index value of zero. It is determined with the help of two types of std oils.

$$V.I = (L - U) / (L - H) \times 100$$

L = Viscosity at 100° Fahrenheit of low viscosity std oil

U = Viscosity at 100° Fahrenheit of the oil under test

H = Viscosity at 100° Fahrenheit of high viscosity std oil

Flashpoint

The flashpoint of an oil is the lowest temperature at which it gives off vapour that will ignite for a moment when a small flame is brought near it. The flash point of a lubricating oil should be greater than 175 degrees Fahrenheit.

The Pensky-Martin closed cup method: it is used to determine the flash point of lubricating oil, fuel oil, solvent, solvent containing material and suspension of solids except cut back asphalt.

Fire point

The fire point of an oil is the lowest temperature at which the vapour of the oil burn continuously for at least 5 seconds when a small flame is brought near it. The flashpoint and fire point are used to indicate the fire hazard of petroleum products and evaporation losses under high temperature operation. Knowledge of flash and fire point in lubricating oil aids in precautionary measures against fire hazards. A good lubricant should have flash point at least above the temperatures at which it is to be used.

Cleveland open cup method can be used to determine the flash and fire point of all petroleum except fuel oils and those having open cup flash below 175-degree Fahrenheit.

Cloud point

The cloud point of petroleum oil is the temperature at which solidifiable compounds, like paraffin wax, present in the oil begin to crystallize or separate from the solution when the oil is cold under specified conditions .At the cloud point oil becomes cloudy or hazy in appearance

Significance

Cloud point is useful for estimating the temperature at which filter screens in the fuel intake system of diesel engines might become clogged because of wax separation .Filterability depends on type of wax, micro crystalline or amorphous .The amorphous wax is sticky in nature and would more easily clog filter screens .To determine cloud point, a sample of lubricating oil is poured into test jar and cooled in progressive steps when the inspection first reveals a distinct cloudiness or haze at the bottom, temperature is recorded as cloud point.

Pour Point

The Pour point of a petroleum oil is the temperature at which the oil ceases to flow or pour. Lubricants used in a machine working at low temperature should possess low Pour point, otherwise solidification of lubrication cause jamming of the machine. Pour point is of importance in establishing lowest temperature at which a diesel fuel is still sufficiently fluid to be pumped or transferred. An oil with low Pour Point should be selected whenever the oil must remain fluid at low temperature.

- Pour point values of petroleum and non-petrol and lubricants are significant as many operations must functions at sub-freezing condition

Aniline point

Aniline point is defined as “the minimum equilibrium solution temperature for equal volume of aniline and oil sample”. It's determined by thoroughly mixing equal volumes of oil sample and aniline in a test tube and heating the mixture until a homogeneous solution is obtained then the tube is allowed to cool at a controlled rate. The temperature at which the oil and aniline phases separate out is recorded as aniline point. A lower aniline point of an oil means a higher percentage of aromatic hydrocarbon are present in it. Higher the percentage of aromatic hydrocarbons or lower the aniline point of an oil, more are the chances of deterioration of an oil when it comes in contact with rubber sealing, packing, etc. Consequently, low aromatic content in the lubricant or their higher aniline point is desirable.